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Program: upGrad and IITB Machine Learning & AI Program | **Course:** SQL and Statistics Essentials

ACPML - Central Limit Theorem - Summary: Central Limit Theorem - Part I - Online

This is a really intense session! Let's summarise everything that's been taught so far and then you can move on the rest of the session.



First, you saw how, instead of finding the mean and standard deviation for the entire population, it is sometimes beneficial to find the **mean** and **standard deviation** for only a small representative **sample**. You may have to do this because of time and/or money constraints.



For example, for an office of 30,000 employees, we wanted to find the average commute time. So, instead of asking all employees, we asked only 100 of them and collected the data. The mean $\bar{X} = 36.6$ minutes and the standard deviation $= \sigma 10$ minutes.

However, it would not be fair to infer that the population mean is exactly equal to the sample mean. This is because the flaws of the sampling process must have led to some error. Hence, the sample mean's value has to be reported with some **margin of error**.



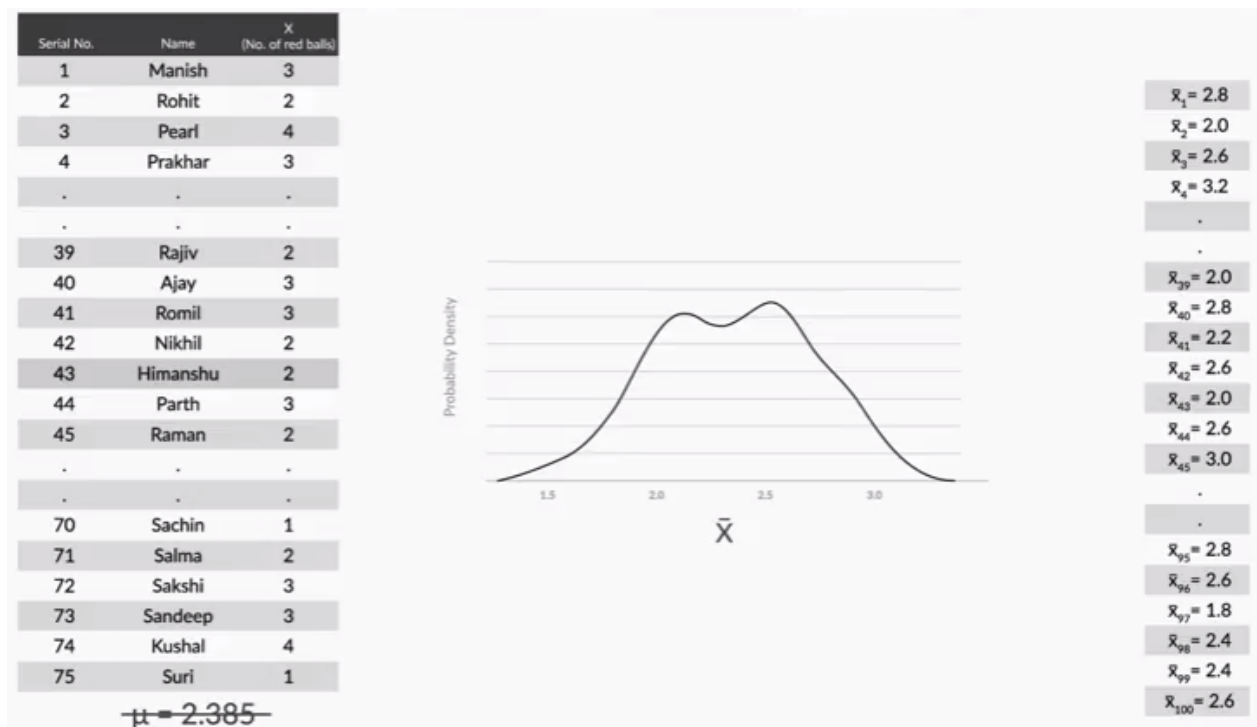
For example, the mean commute time for the office of 30,000 employees would be equal to 36.6 ± 3 minutes, 36.6 ± 1 minutes, or 36.6 ± 10 minutes,



i.e. 36.6 minutes $\pm$ some margin of error.

However, at this point in time, you do not exactly know how to find what this margin of error is.

Then, we moved on to sampling distributions, which have some properties that would help you find this margin of error.



We created a sampling distribution, which was a probability density function for 100 sample means with a sample size = 5.

A sampling distribution, which is essentially the distribution of the sample means of a population, has some interesting properties, which are collectively called the **central limit theorem**. It states that no matter how the original population is distributed, the sampling distribution will follow



these three properties:

1. **Sampling distribution's mean** (μ_{χ}) = **Population mean** (μ)
2. Sampling distribution's standard deviation (**Standard error**) = $\frac{\sigma}{\sqrt{n}}$,
where σ is the population's standard deviation and n is the sample size,
and
3. **For $n > 30$** , the sampling distribution becomes a **normal distribution**.

To verify these properties, we performed sampling using the data collected for our upGrad game from the first session on inferential statistics. The values for the sampling distribution thus created ($\mu_{\chi} = 2.348$, S.E. = 0.4248) were pretty close to the values predicted by theory ($\mu_{\chi} = 2.385$, S.E. = 0.44).

To summarise, the notations and formulas for populations, samples and sampling distributions are as follows:

Population/Sample	Term	Notation	Formula
Population ($X_1, X_2, X_3, \dots, X_N$)	Population Size	N	Number of items/elements in the population
	Population Mean	μ	$\frac{\sum_{i=1}^N X_i}{N}$
	Population Variance	σ^2	$\frac{\sum_{i=1}^N (X_i - \mu)^2}{N}$
Sample ($X_1, X_2, X_3, \dots, X_n$) (Sample of Population)	Sample Size	n	Number of items/elements in the sample
	Sample Mean	$\bar{X}$	$\frac{\sum_{i=1}^n X_i}{n}$
	Sample Variance	S^2	$\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1}$
	Sampling	No convention	



Sampling Distribution of the Sample Mean ($\bar{X}_1, \bar{X}_2, \bar{X}_3, \dots, \bar{X}_k$) (k Sample Means)	Sampling Distribution's Size	NO CONVENTION (We have used k, but that is not a norm)	
	Sampling Distribution's Mean (mean of sample means)	$\mu_{\bar{X}}$	$\mu_{\bar{X}} = \mu$
	Sampling Distribution's Standard Deviation	S.E. (Standard Error)	S.E. = $\sigma/\sqrt{n}$

Now, before moving on to the next lecture, let's spend some time attempting a few practice questions.





